

第2次作业 - 插值和函数逼近

1. 插值

4. 设 $l_0, l_1, \dots, l_n$ 是以节点 $x_0, x_1, \dots, x_n$ 的 n 次 Lagrange 插值基函数, 请证明

$$\sum_{k=0}^n l_k(0) x_k^j = \begin{cases} 0 & , j = 1, 2, \dots, n, \\ (-1)^n x_0 x_1 \cdots x_n & , j = n + 1 \end{cases}$$

【解答】

对于一个 $n+1$ 阶可导的函数 $f(x)$, 根据 Lagrange 插值多项式可知,

$$f(x) = L(x) + R(x) = \sum_{i=0}^n f(x_i) l_i(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i).$$

我们取 $f(x) = x^j$, 对于不同的 j 有如下情况

- 当 $j = 1, 2, \dots, n$ 时, $\forall \xi, f^{(n+1)}(\xi) = 0$, 所以 $x^j = \sum_{k=0}^n f(x_k) l_k(x) + 0$. 取 $x = 0$, 则有 $\sum_{k=0}^n f(x_k) l_k(0) = 0$.
- 当 $j = n + 1$ 时, $\forall \xi, f^{(n+1)}(\xi) = (n+1)!$, 所以 $x^{n+1} = \sum_{k=0}^n f(x_k) l_k(x) + \prod_{i=0}^n (x - x_i)$. 取 $x = 0$, 则有 $\sum_{k=0}^n f(x_k) l_k(0) = (-1)^n x_0 x_1 \cdots x_n$.

综上, 得证

5. 设 $f(x) = x^5 + 4x^3 + 1$, 试求均差 $f[0, 1, 2], f[0, 1, 2, 3, 4, 5], f[0, 1, 2, 3, 4, 5, 6]$.

【解答】

i	x_i	$f(x_i)$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, x_{i+1}, \dots, x_{i+3}]$	$f[x_i, x_{i+1}, \dots, x_{i+4}]$	$f[x_i, x_{i+1}, \dots, x_{i+5}]$	$f[x_i, x_{i+1}, \dots, x_{i+6}]$
0	0	1	$(6-1)/1=5$	$(59-5)/2=27$	$(114-27)/3=29$	$(69-29)/4=10$	1	0
1	1	6	59	114	69	15	1	
2	2	65	287	321	129	20		
3	3	352	929	708	209			
4	4	1281	2345	1335				
5	5	3626	5015					
6	6	8641						

所以 $f[0, 1, 2] = 27, f[0, 1, 2, 3, 4, 5] = 1, f[0, 1, 2, 3, 4, 5, 6] = 0$

6. 给定下表数据, 试构造出 f 的均差表和三次 Newton 插值多项式, 并写出均差型余项

x	1	1.5	0	2
$f(x)$	3	3.25	3	5/3

【解答】

(1) 差商表

i	x_i	$f(x_i)$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, \dots, x_{i+3}]$
0	1	3	$\frac{3.25-3}{1.5-1} = 0.5$	$\frac{1/6-0.5}{0-1} = 1/3$	$\frac{-10/3-1/3}{2-1} = -2$
1	1.5	3.25	$\frac{3-3.25}{0-1.5} = 1/6$	$\frac{-2/3-1/6}{2-1.5} = -10/3$	
2	0	3	$\frac{5/3-3}{2-0} = -2/3$		
3	2	5/3			

(2) 三次牛顿插值多项式

$$\begin{aligned}
 N_3(x) &= f(x_0) + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2) \\
 &= 3 + \frac{1}{2}(x - 1) + \frac{1}{3}(x - 1)(x - \frac{3}{2}) - 2(x - 1)(x - \frac{3}{2})x \\
 &= -2x^3 + \frac{16}{3}x^2 - \frac{10}{3}x + 3
 \end{aligned}$$

(3) 均差型余项

$$\begin{aligned}
 R_3(x) &= f(x) - N_3(x) = f[x_0, x_1, x_2, x_3, x](x - x_0)(x - x_1)(x - x_2)(x - x_3) \\
 &= f[1, 1.5, 0, 2, x](x - 1)(x - 1.5)x(x - 2)
 \end{aligned}$$

12. 求次数不高于3次的多项式 P 使其满足 $P(0) = P'(0) = 0$, $P(1) = 1$, $P(2) = 1$, 并写出其 Newton 形式的余项

【解答】

i	x_i	$f(x_i)$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, \dots, x_{i+3}]$
0	0	0	$P'(0) = 0$	$\frac{1-0}{1-0} = 1$	$\frac{1-(-1/2)}{0-2} = -3/4$
1	0	0	$\frac{1-0}{1-0} = 1$	$\frac{0-1}{2-0} = -1/2$	
2	1	1	$\frac{1-1}{2-1} = 0$		
3	2	1			

所以插值多项式

$$\begin{aligned}
 H_3(x) &= f(x_0) + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2) \\
 &= 0 + 0(x - 0) + 1(x - 0)(x - 0) - \frac{3}{4}(x - 0)(x - 0)(x - 1) \\
 &= -\frac{3}{4}x^3 + \frac{7}{4}x^2
 \end{aligned}$$

Newton形式的余项

$$R_3(x) = f[x_0, x_1, x_2, x_3, x](x - x_0)(x - x_1)(x - x_2)(x - x_3) = f[0, 0, 1, 2, x]x^2(x - 1)(x - 2)$$

13. 求次数不超过4次的多项式 P 使其满足

$P(1) = P(3) = 0$, $P(2) = 1$, $P'(1) = 0$, $P''(1) = 8$ 并写出其 Newton 形式的余项

【解答】

i	x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, \dots, x_{i+3}]$	$f[x_i, \dots, x_{i+4}]$
0	1	0	$P'(1) = \mathbf{0}$	$\frac{P''(1)}{2!} = \mathbf{8/2=4}$	$\frac{1-4}{2-1} = -3$	$\frac{-1-(-3)}{3-1} = 1$
1	1	0	0	$\frac{1-0}{2-1} = 1$	$\frac{-1-1}{3-1} = -1$	
2	1	0	$\frac{1-0}{2-1} = 1$	$\frac{-1-1}{3-1} = -1$		
3	2	1	$\frac{0-1}{3-2} = -1$			
4	3	0				

所以插值多项式

$$H_4(x) = 0 + 0(x-1) + 4(x-1)^2 - 3(x-1)^3 + 1(x-1)^3(x-2) \\ = (x-1)^2(x-3)^2$$

Newton形式余项

$$R_4(x) = f[1, 1, 1, 2, 3, x](x-1)^3(x-2)(x-3)$$

14. 设 $f(x) = \frac{1}{a-x}$, **证明:** **(1)** $f[x_0, x_1, \dots, x_n] = \prod_{j=0}^n (\frac{1}{a-x_j})$, **(2)**

$$\frac{1}{a-x} = \frac{1}{a-x_0} + \dots + \frac{1}{(a-x_0)\dots(a-x_n)}(x-x_0)\dots(x-x_{n-1}) + \frac{1}{(a-x_0)\dots(a-x_n)(a-x)}(x-x_0)\dots(x-x_n)$$

【证明】

(1) 取 x_0, x_1, x_2 , 已知

- $f[x_0, x_1] = (f(x_1) - f(x_0))/(x_1 - x_0) = (\frac{1}{a-x_1} - \frac{1}{a-x_0})/(x_1 - x_0) = \frac{1}{(a-x_0)(a-x_1)} = \prod_{j=0}^1 (\frac{1}{a-x_j})$.
- 同理, 有 $f[x_1, x_2] = (\frac{1}{a-x_2} - \frac{1}{a-x_1})/(x_2 - x_1) = \prod_{j=1}^2 (\frac{1}{a-x_j})$

假设存在恒等式, 对于 $i \geq 0, k > 0, f[x_i, x_{i+1}, \dots, x_{i+k-1}] = \prod_{j=i}^{i+k-1} (\frac{1}{a-x_j})$, 则有

$$f[x_i, x_{i+1}, \dots, x_{i+k}] = \frac{f[x_i, x_{i+1}, \dots, x_{i+k-1}] - f[x_{i+1}, x_{i+2}, \dots, x_{i+k}]}{x_i - x_{i+k}} \\ = \frac{\frac{1}{(a-x_i)(a-x_{i+1})\dots(a-x_{i+k-1})} - \frac{1}{(a-x_{i+1})(a-x_{i+2})\dots(a-x_{i+k})}}{x_i - x_{i+k}} \\ = \frac{1}{(a-x_i)(a-x_{i+1})\dots(a-x_{i+k})} = \prod_{j=i}^{i+k} (\frac{1}{a-x_j})$$

根据归纳法知, 假设成立. 取 $i = 0, k = n$, 即有 $f[x_0, x_1, \dots, x_n] = \prod_{j=0}^n (\frac{1}{a-x_j})$, 证毕

(2) 根据 Newton 插值多项式及其均差形式余项, 再带入第一问结论, 可得

$$\frac{1}{a-x} = f(x_0) + f[x_0, x_1](x-x_0) + \dots + f[x_0, \dots, x_n](x-x_0)\dots(x-x_{n-1}) + f[x_0, \dots, x_n, x](x-x_0)\dots(x-x_n) \\ = \text{RHS}$$

15. 设 $f(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1} + a_nx^n$ **有** n **个不同的实根** $x_1, x_2, \dots, x_n$, **试证明**

$$\sum_{j=1}^n \frac{x_j^k}{f'(x_j)} = \begin{cases} 0 & , 0 \leq k \leq n-2 \\ a_n^{-1} & , k = n-1 \end{cases}$$

【解答】

改写为 $f(x) = a_n(x-x_1)(x-x_2)\cdots(x-x_n)$, $f'(x) = a_n \sum_{i=1}^n \prod_{j \neq i} (x-x_j)$, 取 $x = x_i$ 则有

$$f'(x_i) = a_n \prod_{j \neq i} (x_i - x_j).$$

$$\text{所以 } l_i(x) = \prod_{\substack{j=1 \\ j \neq i}}^n \frac{(x-x_j)}{(x_i-x_j)} = \frac{f(x)/(x-x_i)}{f'(x_i)} = \frac{f(x)}{(x-x_i)f'(x_i)}.$$

取多项式 $P(x) = x^k, 0 \leq k \leq n-1$, 有 $P^{(n)}(x) = 0$. 取 $\{x_1, x_2, \dots, x_n\}$, 做 $n-1$ 次Lagrange多项式插值得

$$\begin{aligned} x^k = P(x) &= \sum_{i=1}^n P(x_i)l_i(x) + \frac{P^{(n)}(\xi)}{(n+1)!} \prod_{i=0}^n (x-x_i) \\ \iff x^k &= \sum_{i=1}^n \frac{x_i^k f(x)}{(x-x_i)f'(x_i)} = \sum_{i=1}^n \frac{x_i^k}{f'(x_i)} \frac{f(x)}{(x-x_i)} \end{aligned}$$

考察恒等式两侧最高次项系数: 已知项 $\frac{f(x)}{(x-x_i)}$ 是一个 $n-1$ 次多项式, 其最高次项系数为 a_n , 因此右侧多项式的最高次项系数为 $\sum_{i=1}^n \frac{a_n x_i^k}{f'(x_i)}$, 次数 $n-1$; 左侧多项式最高次项系数为 1, 次数为 k :

- 若 $k = n-1$, 则第 $n-1$ 阶系数有 $\sum_{i=1}^n \frac{a_n x_i^k}{f'(x_i)} = 1$.
- 若 $0 \leq k \leq n-2$, 则第 $n-1$ 阶系数有 $\sum_{i=1}^n \frac{a_n x_i^k}{f'(x_i)} = 0$.

合并两种情况, 即得证

2. 函数逼近

7. 设 $f(x) = e^x, x \in [-1, 1]$, 试求出 f 在 $\mathcal{P}_1$ 和 $\mathcal{P}_2$ 中的最佳平方逼近多项式 P_1^* 和 P_2^* .

【解答】

在 $\mathcal{P}_n$ 中, 基函数 $\phi_0 = 1, \phi_1 = x, \phi_2 = x^2$. 相互之间的内积如下

$$\begin{aligned} (\phi_0, \phi_0) &= \int_{-1}^1 1 dx = 2, (\phi_0, \phi_1) = \int_{-1}^1 x dx = 0, (\phi_0, \phi_2) = \int_{-1}^1 x^2 dx = \frac{2}{3} \\ (\phi_1, \phi_1) &= \int_{-1}^1 x^2 dx = \frac{2}{3}, (\phi_1, \phi_2) = \int_{-1}^1 x^3 dx = 0 \\ (\phi_2, \phi_2) &= \int_{-1}^1 x^4 dx = \frac{2}{5} \end{aligned}$$

与 $f(x)$ 的内积如下

$$(f, \phi_0) = \int_{-1}^1 e^x dx = e - e^{-1}, (f, \phi_1) = \int_{-1}^1 x e^x dx = 2e^{-1}, (f, \phi_2) = \int_{-1}^1 x^2 e^x dx = e - 5e^{-1}$$

所以, 对于 $P_1^*(x) \in \mathcal{P}_1$, 有

$$\begin{aligned} \begin{bmatrix} 2 & 0 \\ 0 & \frac{2}{3} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} &= \begin{bmatrix} e - e^{-1} \\ 2e^{-1} \end{bmatrix} \\ a_0 &= \frac{e - e^{-1}}{2}, a_1 = 3e^{-1} \\ P_1^*(x) &= \frac{e - e^{-1}}{2} + 3e^{-1}x \end{aligned}$$

对于 $P_2^*(x) \in \mathcal{P}_2$, 有

$$\begin{aligned} \begin{bmatrix} 2 & 0 & \frac{2}{3} \\ 0 & \frac{2}{3} & 0 \\ \frac{2}{3} & 0 & \frac{2}{5} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} e - e^{-1} \\ 2e^{-1} \\ e - 5e^{-1} \end{bmatrix} \\ \text{解得 } a_0, a_1, a_2 \text{ 带入} \\ P_2^*(x) &= \frac{3(11 - e^2)}{4e} + 3e^{-1}x + \frac{15}{4}(e - 7e^{-1})x^2 \end{aligned}$$

9. 设 $f(x) = x^2 + 3x + 2, x \in [0, 1]$, 试求出 f 在 $\mathcal{P}_2$ 中的最佳平方逼近多项式 P_2^* .

【解答】

$f \in \mathcal{P}_2$, 其在 $L^2[0, 1]$ 上对 $\mathcal{P}_2$ 的正交投影即为自身

所以 $P_2^*(x) = f(x) = x^2 + 3x + 2$

12. 已知数据如下表, 试用 $y = ax^2 + bx + c$ 来拟合上述数据

x_i	-3	-1	1	3
y_i	15	5	1	5

【解答】

即用多项式进行最小二乘曲线拟合. 记向量 $\phi_i = [\cdots, \phi_i(x_i), \cdots]$, $f = [\cdots, f(x_i), \cdots]$

$$\begin{aligned}(\phi_0, \phi_0) &= \sum 1 = 4, (\phi_0, \phi_1) = \sum x_i = 0, (\phi_0, \phi_2) = \sum x_i^2 = 20 = (\phi_1, \phi_1) \\(\phi_1, \phi_2) &= \sum x_i^3 = 0, (\phi_2, \phi_2) = \sum x_i^4 = 164 \\(f, \phi_0) &= \sum y_i = 26, (f, \phi_1) = \sum x_i y_i = -34, (f, \phi_2) = \sum x_i^2 y_i = 186\end{aligned}$$

所以可得

$$\begin{bmatrix} 4 & 0 & 20 \\ 0 & 20 & 0 \\ 20 & 0 & 164 \end{bmatrix} \begin{bmatrix} c \\ b \\ a \end{bmatrix} = \begin{bmatrix} 26 \\ -34 \\ 186 \end{bmatrix}$$

$$\therefore a = \frac{7}{8}, b = -\frac{17}{10}, c = \frac{17}{8}$$

因此 $y = \frac{7}{8}x^2 - \frac{17}{10}x + \frac{17}{8}$

补充: 求函数 $f(x) = |x|$ 在区间 $[-1, 2]$ 上的最佳一致一次逼近多项式

观察 $f(x) = |x|$, 选择区间 $[-1, 2]$ 两侧端点和最小值折点作为交错点组 $\{-1, 0, 2\}$

所以多项式 $g(x) = ax + b$ 应满足, 记最小偏差为 E

$$\begin{aligned}f(-1) - g(-1) &= 1 + a - b = E \\g(0) - f(0) &= b = E \\f(2) - g(2) &= 2 - 2a - b = E\end{aligned}$$

解得 $a = \frac{1}{3}, b = E = \frac{2}{3}$. 所以 $g(x) = \frac{1}{3}x + \frac{2}{3}$.

经验证 $E(x) = ||x| - g(x)|$ 在区间上最大值为 $\frac{2}{3}$, 仅在交错点组上取到, 符合要求. 根据Chebyshev交替定理, 最佳一致一次逼近多项式 $P_1^*(x) = g(x)$